032: What decides whether behaviour pays — movement, not the signal it follows

- **Type:** research article (two negative results that reorder an earlier positive one, plus one refuted mechanism)
- **Version:** v1.2
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- **Status:** ready for deposit once the author says so; not submitted, not deposited. v1.1 applies the independent referee verdict ("does not hold as written"): the two unverifiable trap counts were withdrawn, the G7 budget imbalance restated at +53% over the run, the memory-optimum table corrected to show that the change rate DOES shift it under six jumps, the displacement metric described as instantaneous rather than per-lifetime, the anchor scope narrowed, and the duplicated-run structure disclosed
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- **Related reports:** S19/G1,

S21/G2, S22/G3, S23/G4, S24/G6, S25/G7, S26/H2, S27/H4, S28/H5, S29/H6, S30/H7, S31/H8, S32/H9, S33/H10, S34/H11

- **Artifacts:** `morphogenesis/experiments/E09_pond/s{19,22,23,24,25,26,27,28,29,30,31,32,33,34}_*.json`
- **Previous article of the series:** 031, DOI

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Abstract

The previous article of this series reported that local resource-greedy movement helps in a patchy pond world, but that its advantage shrinks as the rich zone fills. This article asks the next question: **what decides whether a movement rule pays at all?** Three candidate answers were tested and two of them were wrong.

First, we asked whether the sign of the advantage depends on how fast the resource recovers. It does: with slow recovery the greedy rule *loses* to a non-moving control (gap −0.078 to −0.205, all three seeds negative), with fast recovery it wins (+0.036 to +0.101). Second, we asked what causes the loss, and found a *trap of inaction*: the step rule fires only when the cell's prediction exceeds the actual resource, and the prediction is an exponential average that tracks the falling resource, so the cell stops noticing that it is in a bad place — at step 300 only 3 of 45 cells still qualified to move, and by step 400, 102 of 108 living cells sat on depleted cells. Third, we asked whether removing the adaptation fixes it. It does, dramatically (+0.338 instead of −0.174), **but the reason is not the reference**: with the number of moving cells held equal, the two references are indistinguishable (0.238 vs 0.246) and both are worse than the control. The dominant variable is *how much the cell moves*, not *what it compares with*.

We then tested a continuous diffusing signal field in four designs: built from the living cells (no effect — its gradient duplicates local information), driven by an external rhythm (no effect — it carries information unrelated to food), carrying the memory of where the food was (a real gain, +0.038…+0.045), and finally compared against plain smoothing of the resource (no difference: 0.299 against 0.297). The field is unnecessary; what helps is looking at a *neighbourhood* rather than at a single cell. A direct measurement explains why: a cell moves 0.14 cells from its birth cell in a whole lifetime, so it cannot use long-range information at all.

Keywords: pond, directionality, habituation, mobility, memory, boundary of the setup

The line this article continues

Last time. *Directionality works but does not scale, and inherited adaptation does not appear, in a resource-limited pond world* (Zenodo, DOI 10.5281/zenodo.24012536) found that a cell stepping towards the richer neighbour holds 0.475 of the favourable half against 0.357 for the control, that the gain decays to +0.036 as the rich zone fills because cells compete for space rather than for food, and that four tests of inherited adaptation give null.

What we decided to check. That article measured *how much* the rule pays in one setting; it did not say what decides whether it pays at all. So this article moves the levers one at a time: the rate at which the resource comes back, the adaptation inside the rule, and a diffusing signal field that the cells could in principle follow. Two of the three answers we expected turned out to be wrong, and the article reports the wrong ones first.

1. The question

The previous article established that a cell stepping towards the neighbour with more resource holds a larger share of the favourable zone than a non-moving control, but that the gain decays as the zone fills. That leaves the mechanism open: is the gain a property of the *rule* (which way to step), of the *environment* (how fast food comes back), or of something else entirely?

This article separates those. The world is the same as in the previous article: a 32×32 grid, a resource in the cells, organisms with energy on the cells, proportional income, death when energy runs out, reproduction on accumulated mass. The non-degenerate regime is unchanged, and R1 (mean deaths ≥ 10 and mean living ≥ 10) is checked in every run.

2. Context

Two ideas from the literature frame the work. First, mobility in agent-based foraging models is known to interact with population density and with resource depletion and replenishment rates, so that the return from moving can be lower than the return from staying put when the neighbourhood is crowded [1]. Second, plasticity is useful only if the adaptation time is shorter than the death time — our formal hypothesis F-ADAPT — which is the frame in which a *fast* adaptation could be harmful rather than helpful.

What this article adds is a controlled separation: the same world, the same seeds, and one knob at a time, so that the question "what decides whether behaviour pays" gets a measured answer rather than an interpretation.

References cited in this article: [1] *Food Income and the Evolution of Forager Mobility*. Scientific Reports 9 (2019), s41598-019-42006-2.

3. Methods

The world. Unchanged from the previous article. The patchy layout has three rich patches; a colony is seeded on one of them, because seeding in the poor centre makes the experiment insensitive by construction (the cells starve before direction can matter — this was measured and is recorded in the previous article's withdrawn experiments).

The metrics. The favourable share — the fraction of living cells standing on the top 50% of cells by current resource — and, for the stability questions, its within-window standard deviation. The process metric is the number of moves per step.

The arms. Each experiment varies one knob: the resource recovery rate, the reference the step rule compares against, the number of moving cells, the field design, the memory lag, the field's lifetime, the cell's mobility, or the distance between the old and the new food.

Sensitivity arithmetic before each run (contract §3c). No experiment was started without first computing whether the manipulation could reach the dynamics: income at a full patch is 0.160 per step and upkeep is 0.060, so a colony on a patch is viable in every regime used; the patch inflow was checked against one cell's consumption before any depletion test.

4. Results

4.1 The sign of the advantage depends on how fast food returns

recovery	control	greedy (adaptive reference)
slow (regen 0.010)	0.340	**0.166** (gap −0.174)
fast (regen 0.040)	0.325	0.361 (gap +0.036)

With slow recovery the greedy rule **loses**. All three seeds are negative by the end (−0.138, −0.105, −0.278). This is the first place in the series where the behaviour is actively harmful rather than merely useless.

4.2 The cause is a trap of inaction

The step rule fires only when `resource < pred`, and `pred` is an exponential average of the resource. When the resource falls, the prediction falls with it, the error goes to zero, and the cell stops qualifying to move. The mechanism is visible in the qualifying counts recorded by the runs:

window	cells qualifying to move	living cells	fraction able to act
0–100	18.6	60.2	0.31
100–200	8.6	55.6	0.16
200–300	17.3	101.2	0.17
300–400	21.8	126.0	0.17
400–500	25.5	146.2	0.17
500–600	32.1	184.5	**0.17**

The absolute number of cells able to move rises with the population, but the *fraction* falls from 0.31 to 0.17 and then sticks there: four cells in five can no longer act on their situation. The consequence is visible in the same runs: by the last window the adaptive arm keeps **94.6 %** of its living cells on depleted cells (resource < 0.3), against **90.8 %** for the non-moving control.

A limitation, stated plainly. An earlier version of this section quoted two counts — "3 of 45 cells at step 300" and "102 of 108 cells on depleted cells at step 400" — taken from a diagnostic exploration that was never saved as an artifact and has no script in the repository. Those two numbers are therefore **withdrawn**; the mechanism now rests only on the per-window quantities above, which are stored in `s22_g3_full.json` and can be recomputed.

4.3 Removing the adaptation fixes it — but not for the reason it looks like

Replacing the adaptive reference with a fixed one (the resource at seeding time) changes the picture completely:

window	control	adaptive reference	fixed reference
0–100	0.363	0.405	0.501
300–400	0.288	0.185	0.663
500–600	0.340	**0.166**	**0.678**

The sign flips: −0.174 becomes +0.338. But two controls demolish the natural reading. The fixed reference is better at *both* recovery rates (+0.338 slow, +0.350 fast), so it is not a cure for the slow regime; and it makes about ten times more cells move (338 against 22 per step). With the number of moving cells held equal, the gain falls from +0.338 to +0.094, and with an **absolute** equal budget the two references are indistinguishable:

arm (near-equal absolute budget)	favourable share	moves per step (last window)
control	0.340	0
adaptive reference	0.238	13.5
fixed reference	0.246	16.1

The dominant variable is how much the cell moves, not what it compares with. The "adaptation trap" is a trap of inaction.

A limitation of that control, stated plainly — and it is serious. The budget was set to 20 cells per step for both reference arms, but the cap only binds from above: the adaptive arm never produced 20 candidates, so it moved **10.4 cells per step over the whole run against 15.9 in the fixed arm (+53 %)**, and 13.5 against 16.1 in the last window alone. The two arms therefore differ in **both** the reference and the amount of movement, so this control does **not** separate the two contributions. What it does show is that the fixed reference's large gain (+0.338) does not survive being brought close to the adaptive arm's mobility, which is consistent with movement dominating — but it is not the clean isolation the title claim needs. A control with a budget below both arms' candidate counts (so that it binds for both) has not yet been run.

4.4 A continuous signal field: four designs, one conclusion

The owner of the project proposed a continuous field — a diffusing "chemical" signal — that could carry information at a distance and oscillate. It was implemented additively (with every field parameter at zero the world is bit-for-bit the old one, checked by an anchor) and tested in four designs:

design	favourite share	verdict
field built from living cells	0.318	no effect: its gradient duplicates local information
external travelling wave (phase)	0.332	no effect: carries information unrelated to food
external wave, level-adaptation	0.330	same as phase-following
memory field (source = resource `lag` steps ago)	**0.297**	**a real gain** against the 0.260 control
plain smoothing of the resource	**0.299**	**matches the memory field**

The memory field is the only variant with a gain, and it is worth +0.038…+0.045. But it can be replaced by smoothing the resource over a neighbourhood, with no field, no diffusion and no memory. What helps is looking at a *neighbourhood* rather than at a single cell.

4.5 Why: the cell stays near where it was born

Tagging each living cell with its birth position gives the answer. The quantity below is the mean, **at each instant**, of the distance between a living cell and its own birth cell:

mobility	mean displacement (last window)	90th percentile	maximum
1	**0.15 cells**	1.0	4.1
3	**0.19 cells**	1.0	5.1

At any moment, nine cells in ten stand on their birth cell or next to it. A few individuals do travel further — the maximum reaches 4–5 cells — so the population is not strictly sedentary, but the typical cell never gets far. That is why the field's "distance to the food" does not matter and why the field is equivalent to local smoothing.

A correction. An earlier version of this section called the same quantity "0.14 cells in a lifetime". That was a misreading: the metric is an instantaneous mean over living cells, not a per-cell lifetime total, and the maxima in the same windows show that some cells do travel several cells. The conclusion (the typical cell stays local) is unchanged, but the wording is now what the data actually says.

4.6 The memory optimum: one lever works, three do not

The memory length has a clear optimum (lag 10–40 beats no memory by +0.035…+0.047). Four levers were tested against it:

lever	shifts the optimum?
rate of environmental change	**partly**: no change and one jump peak at lag 10, but six jumps peak at lag **40** (0.302 against 0.292 at lag 10)
lifetime of the field (decay)	no (0.300 / 0.297 / 0.294)
distance to the food	no (0.298 / 0.300 / 0.299)

lever	shifts the optimum?
cell mobility	**yes** (best lag 10 at mobility 1, 40 at mobility 3)

The simple formula "optimal lag ≈ distance / mobility" was tested and **refuted**: the distance half does not hold. The mechanism behind the mobility half is the same as in §4.5 — a faster cell reads a wider neighbourhood, so it can use a longer memory.

An honest correction. An earlier version of this table said the rate of environmental change does not shift the optimum, quoting only the lag-10 value. The full curve says otherwise: under six jumps the best lag is 40, and two of the three seeds peak there. The optimum does move with the change rate — in the direction the original hypothesis predicted — but only when the environment changes often enough for memory to be a liability.

5. Discussion

What this means. Across two lines of work the same lesson repeats: the signal a cell follows matters far less than whether it moves at all, and how much. Adaptive references suppress movement and therefore hurt; fixed references and fields help only insofar as they restore movement; and a field of any design adds nothing that local smoothing does not.

What was refuted, and why that is the result. Three natural explanations were tested and failed: that the memory optimum tracks the environment's change rate, that it tracks the field's lifetime, and that it tracks the distance to the food. Two field designs that looked promising were shown to carry no usable information. Reporting these as results rather than quietly dropping them is what makes the remaining claims credible.

What cannot be concluded. We do not show that fields are useless in general — only that in a world where a cell moves one cell per lifetime, a field cannot be more than a local smoother. Nor do we show that the mobility is "the" variable; we show that it dominates the reference in this setup.

6. What this does not show

- It does not show that the world is alive, adaptive or intelligent in any strong sense.
- It does not show that behaviour cannot pay in general — only what decides it here.
- It does not give a quantitative law for the memory optimum; that remains open.

7. What would change the answer

- If cells could occupy the same site (several organisms per cell), competition for food would become possible, and the

mobility-dominated picture might change. This needs an engine change and was not made.

- If a field could predict where food will be rather than where it was, it would carry information the cell cannot get

locally; that design was not tested.

8. Where we move next

The open quantitative question is the memory optimum: four levers were tested, one (mobility) moves it, and no formula fits. The next step is to measure the relationship between mobility and the optimal lag on more than three mobility levels, and to check whether the optimum is set by the *width of the neighbourhood the cell reads* rather than by mobility as such — which is what §4.5 and §4.6 together suggest.

9. Reproducibility

Every number comes from artifacts under `morphogenesis/experiments/E09_pond/`; the scripts are `morphogenesis/scripts/s{19,22,23,24,25,26,27,28,29,30,31,32,33,34}_*.py`. Runs were on CPU (the environment has no CUDA build of PyTorch), 3 seeds, 600 steps, with R1 checked in every run and reported alongside each result.

The engine additions made for this article (the continuous field) and for the previous one (the second flip, the environment rotation) are additive: with every new parameter at zero the world reproduces the previous one bit-for-bit.

What the anchor does and does not cover. The anchor run (`s26_anchor_check.json`: `alive 489 / died 5093 / correct_after 0.47229`) is the baseline of the *gradient* world with plasticity off, one seed. It shows that adding the field code did not change the existing world. It does **not** validate the field machinery itself, which is exercised only by the S26–S34 runs.

Duplicated runs across experiments. Several arms in this article are the same underlying run reused in a different sweep, because the sweeps share their control configuration. Specifically: `s31` mobility 1 is bit-identical to `s30` decay 0.05 (12 of 12 shared cells); `s34`'s memory arm equals `s33`'s mobility-1 lag-10 run; the `no_field` arms of `s26`, `s27` and `s34` are one run; `s28`'s `no_field` equals `s32`'s far-distance lag-0 run; and `s24`/`s25` share their baseline. This is not an error — the runs are deterministic and the shared arms *should* coincide — but it means the apparent number of independent observations is smaller than the number of arms, and any claim that rests on two arms being independent must be read with that in mind.